

Jinchen Zhao

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EDUCATION	Yale University , New Haven, CT Ph.D. Student, Applied Physics	Aug 2023 – Present
	Duke Kunshan University , Kunshan, China Duke University , Durham, NC B.S., Dual-degree, Applied Mathematics	2019 – 2023
RESEARCH INTERESTS	Quantum Information, Deep Learning, Quantum Optics. Quantum-Enhanced Physical Neural Networks, Nonlinear Photonics, Quantum Phase Transitions.	
RESEARCH EXPERIENCE	<i>Graduate Research Assistant</i> , Yale University Advisor: Dr. Logan Wright	Aug 2023 – Present
	Studying unconventional computing platforms (optical, quantum, and thermodynamic), focusing on the role of analog noise in computation at ultra-low energy scales and pathways toward practical, scalable applications.	
	<i>Undergraduate Research Assistant</i> , Duke Kunshan University Advisor: Dr. Myung-Joong Hwang	2021 – 2023
	Discovered the frustration of photons during the superradiant phase transition of a Dicke lattice model and studied its novel critical behaviors.	
	<i>Undergraduate Research Assistant</i> , Duke University Advisor: Dr. Kenneth Brown	Spring 2022
	Developed an auto-loading module for the trapped ion system, which tested successfully on the device and was added to the official control system.	
	<i>Undergraduate Research Assistant</i> , Duke Kunshan University Advisor: Dr. habil. Kai Huang	2019 – 2021
	Monitored projectile motion under opaque granular material by embedded inertial measurement unit, and studied granular impact under micro-gravity.	
PUBLICATIONS	[1] Jinchen Zhao and Logan Wright, <i>Quantum-Enhanced Analog Computing</i>, in preparation. [2] Tivadar Pongó*, Tianhui Liao*, Jinchen Zhao, Valentin Dichtl, Simeon Völkel, Raúl Cruz Hidalgo, and Kai Huang, <i>Nature of granular drag in microgravity</i>, arXiv:2603.03006, under review at <i>Phys. Rev. Lett.</i> [3] Jinchen Zhao and Myung-Joong Hwang, <i>Anomalous criticality with bounded fluctuations and long-range frustration induced by broken time-reversal symmetry</i>, <i>Phys. Rev. Research</i> 5, L042016 (2023). [4] Jinchen Zhao and Myung-Joong Hwang, <i>Frustrated Superradiant Phase Transition</i>, <i>Phys. Rev. Lett.</i> 128, 163601 (2022). [5] Stefan Köstler, Jinchen Zhao, Chen Lyu, Simeon Völkel and Kai Huang, <i>Embedded Inertial Sensor for Tracking Projectile Impact on Granular Media</i>, EPJ	

PRESENTATIONS	[1] <i>Anomalous Multicritical Phenomena and Frustration Induced by Synthetic Magnetic Fields</i> , APS March Meeting, 2023.	
INDUSTRY EXPERIENCE	<i>Research Intern</i>, NTT Research, Physics & Informatics (PHI) Lab 2026 - Studied the role of quantum optics in optical computing, and tensor computing in programmable nonlinear multimode waveguides. <i>University-Corporation Lab</i>, World Economic Forum 2022 - Led a team of 5 students to map the East Asian quantum computing ecosystem, synthesizing policy, industry, and academic data for the WEF Technology and Quantum Computing Topic Leader.	
HONORS	Optics and Photonics Scholarship – SPIE 2026 Jane Street PhD Fellowship Honorable Mention – Jane Street 2026 Roberts Innovation Fund – Yale University 2025 Sterling Prize Fellowship – Yale University 2023 Zu Chongzhi Math Student First Award – Duke Kunshan University 2023 <i>Magna Cum Laude</i> Latin Honor – Duke Kunshan University 2023 Graduation with Distinction – Duke Kunshan University 2023	
SERVICE	Advisor of 5 undergraduate research projects, Yale University Reviewer, <i>Nature Photonics</i> Reviewer, <i>Nature Communications</i> Reviewer, <i>Light: Science & Applications</i> Reviewer, <i>Journal of the Optical Society of America B</i> Teaching Fellow, APHY 5750 Physics of AI, Yale University Teaching Fellow, PHYS 180 University Physics, Yale University	
SKILLS	Programming: Python (PyTorch, QuTiP, Qiskit), Verilog/HDL Hardware: RFSoc/FPGA (Vivado), RF & photonic instrumentation control Tools: Mathematica, Git, L^AT_EX	